

Work Experience Module 1 - Student Enrollment System

Technical Requirement Analysis and Refinement

Module #: 251201-005-00-WM-01

NQF Level: 5

Notional hours: 150

Credit(s): 15

Curriculum Code: 251201005

Qualification Title: Occupational Certificate: Software Developer

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Introduction

As part of your training you are required to keep a logbook of all practical on the job training and exposure you receive during the learning process.

You will now be assigned to a mentor who will oversee your off-site training, usually referred to as on-the-job training.

The mentor will assist and advise you on the practical aspects of the job, how to fit into the company, what is expected of you as an employee and as a future supervisor.

Responsibilities of the learner include:

- One hundred percent commitment to the learning process. Learners are encouraged to study any additional source of information relevant to this learning process.
- Doing all assignments contained in this workbook as well any tasks and assignments received from your mentor or supervisor to whom you have been assigned.
- Although the mentor is responsible to sign off all sections completed, it is the learner's responsibility to ensure that all paperwork is completed and handed in for filing on his/her record of learning. It should be clearly stated to learners that a 100% complete record of learning, as prescribed by this logbook, is **their sole responsibility**. Any document missing from the record may result in your not being declared competent.
- Discuss any problems that you may have with your mentor.

Acknowledgment of Receipt

I **Sashin Singh** (Learner) acknowledge receipt of my on the job Training Logbook on this the **15th** day of **January** 2025.

The process of on-the-job training has been explained to me.

Signature of Learner: S. Singh

Name of Mentor: Allen Makandla

Signature of Mentor: A. Makandla

Declaration of Authenticity

I (learner name and surname) **Sashin Singh**, ID Nr **9501125487082** hereby declare that the work contained herein was completed by me on my own.

Where assistance or advice was received or where I used resource material from a workbook, policy wording, internet, or any other printed sources, this has been acknowledged and referenced. I further declare that I understand that plagiarism is a punishable offence as it constitutes the theft of another's intellectual property rights.

Learner signature: S. Singh **Date:** 15/01/2025

Declaration by mentor

I (mentor name and surname) **Allen Makandla**, ID Nr **8703145698741** hereby declare that the learner is being mentored by myself and that the functions listed and the working hours is a true reflection of the learner's situation. According to my knowledge I declare that this is his/her own work.

Instructions to the Mentor

Congratulations on your appointment as a mentor to the learner. This is a very responsible assignment because you have been tasked with the responsibility of rounding off the learner's practical exposure.

You must ensure that you are familiar with all aspects of the work covered in this logbook because you must keep a daily account of the learner's performance.

You are also required to report to the learnership coordinator skills development facilitator, or as agreed between yourself, the coordinator and the learner regarding the learner's progress. Your responsibilities as mentor are as follows:

- Study the logbook and acquaint yourself with its content and format
- Remember this logbook is the learner's full record of learning and workplace exposure. All activities which the learner participates in must be recorded, and all documents produced in relation to this learnership must form part of the record of learning
- Get all the learners together and explain its purpose to them and also what is required of them
- Remember the mentor is the creator of learning and exposure opportunities. You should therefore not confine the learner's exposure to this logbook alone
- Continuously guide them in doing the assignments and arranging the planned exposure with the relevant departments
- File all duplicate records of learning on a file for each learner
- Send all original records to the training provider at the end of each month. The Seta also requires that copies be held at the companies
- This logbook need not be followed chronologically, but please note that the learner is required to work through the entire logbook by the end of the learning period

Responsibilities of the Employer

- Creating an atmosphere conducive to learning
- Giving learners ample access to the working environment. Remember that learners should be productive employees to get practical exposure to all aspects of the software development operation as required by the learnership
- Ensure that learners, mentors and assessors attend all training required and arrange and pay travelling and accommodation costs
- Ensure availability of sufficient mentors and workplace assessors

Training Provider Responsibility

- Provide all practical learning material in electronic and/or hard copy, depending on the circumstances
- Provide training for mentors and workplace assessors if required and provide the learning material

- Visit employers to monitor progress and provide guidance and feedback
- Provide an online and telephonic support system to all mentors and learners

Provider Accreditation Requirements for the Module

Provider Accreditation Requirements:

Physical Requirements:

Demonstrate access to:

- Development tools and IDEs
- Work space and computer per learner
- Internet connectivity and development servers
- Standard operating procedures
- Software documentation and specifications

Human Resource Requirements:

- Qualified software developer with 3 years of experience for every 2 learners
- Recognised expertise in mentorship
- Recognised expertise in assessment

Legal Requirements:

- Compliance to the relevant occupational health, safety and environmental protection legislation

Additional Assignments to be Assessed Externally None specified

WORK EXPERIENCE MODULE SPECIFICATIONS

This qualification also requires the following Work Experience Modules:

List of Work Experience Module Specifications:

- 251201005-WM-01, Technical Requirement Analysis and Refinement, NQF Level 5, Credits 15
- 251201005-WM-02, Modelling Processes, NQF Level 5, Credits 15
- 251201005-WM-03, Programming for Software Solution Development, NQF Level 5, Credits 25
- 251201005-WM-04, Capstone project, NQF Level 5, Credits 10

Details of Work Experience Module 1

The focus of the work experience is on providing the learner an opportunity to:

Interrogate the specification and problem, interpret it into code and articulation in writing.

The learner will be required to:

- WM-01-WE01 : Attend induction program and familiarise self with company processes, procedures, tools and culture.
 - WM-01-WE02 : Observe and assist a software engineer with gathering of information and analysis of the requirements and problems.
 - WM-01-WE03 : Under supervision, gather information and analyse the requirements and problems by conducting the following tasks.
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WM-01-WE01: Attend induction program and familiarise self with company processes, procedures, tools and culture

Scope of Work Experience

The person will be expected to engage in the following work activities:

- WA0101 Attend induction program and familiarise self with the culture of the company
- WA0102 Familiarise self with legislation in the workplace
- WA0103 Apply protocols and work etiquette
- WA0104 Attend company specific information sharing sessions (e.g. standing meetings, toolbox talks, power hours, etc.)
- WA0105 Familiarise self with and apply "working from anywhere" protocols
- WA0106 Comply with governance protocols and code of ethics of the company and ensure legal compliance by adhering to legal requirements (incl. but not limited to privacy, confidentiality, security of information and business intelligence, etc.)
- WA0107 Spend time in the various departments of the company, observe process flows and do modelling of the processes observed using suitable tools and showing the relationships and influences the respective departments have on each other
- WA0108 Understand management requirements and expectations from software development and software solutions
- WA0109 Understand business computer system and the support required from the software development team
- WA0110 Manage timesheets and apply self-management skills
- WA0111 Collaborate with team members throughout the work experience period to achieve common and individual goals

Supporting Evidence

- SE0101 Attendance register

WM-01-WE01 ATTEND INDUCTION PROGRAM AND FAMILIARISE SELF WITH COMPANY PROCESSES, PROCEDURES, TOOLS AND CULTURE

Item	Scope	Comments	Date	Signature
WA0101	Attend induction program and familiarize self with the culture of the company: Example: Attend the company's orientation sessions where you learn about the company's mission, values, and the web development culture. Participate in team-building activities designed to immerse new employees in the company's collaborative and innovative work environment focused on HTML/CSS/PHP/JavaScript development.	Completed 5-day comprehensive induction covering company values, web development standards, coding practices, and team collaboration. Participated in orientation activities and technical onboarding sessions.	15-19 Jan 2025	A. Makandla
WA0102	Familiarize self with legislation in the workplace: Example: Study the specific laws and regulations that affect your work, such as data protection laws (GDPR, POPIA), software licensing, intellectual property rights, and web accessibility standards to ensure you handle data and software assets legally.	Studied workplace legislation including GDPR compliance, intellectual property laws, web accessibility standards (WCAG 2.1), and software licensing. Completed compliance quiz with 96% score.	19-Jan-2025	A. Makandla
WA0103	Apply protocols and work etiquette: Example: Adheres to the company's code review guidelines for PHP and JavaScript, regularly participates in pair programming sessions, and follows the established communication protocols for reporting progress and setbacks in daily stand-up meetings.	Implemented Git workflow protocols, participated in code review processes, established proper communication channels for project updates, and adhered to coding standards for PHP/HTML/CSS/JavaScript development.	16-Jan-2025	A. Makandla
WA0104	Attend company specific information sharing sessions: Example: Attends weekly "tech talks" where team members share knowledge on new web technologies, PHP frameworks, JavaScript libraries, project updates, and coding best practices, fostering a culture of learning and continuous improvement.	Participated in "Modern Web Development" tech talk, contributed to discussion on PHP 8 features and JavaScript ES6+ implementations, attended MySQL optimization workshop.	17-Jan-2025	A. Makandla

Item	Scope	Comments	Date	Signature
WA0105	Familiarize self with and apply "working from anywhere" protocols: Example: Sets up your home office according to the company's remote work guidelines, ensuring you have secure VPN access, properly configured XAMPP development environment, follows prescribed online communication practices, and maintains productivity while working remotely.	Configured secure home development environment including XAMPP server, established VPN connection, set up development tools (VS Code, Git), implemented security protocols for remote work.	19-Jan-2025	A. Makandla
WA0106	Comply with governance protocols and code of ethics of the company: Example: Ensures all code is documented and meets the company's standards for quality and security. They adhere to ethical guidelines around data handling, user privacy, secure coding practices for web applications, ensuring transparency and fairness in software solutions.	Signed comprehensive code of ethics agreement, implemented secure coding practices including SQL injection prevention, XSS protection, and proper data validation in PHP applications.	19-Jan-2025	A. Makandla
WA0107	Spend time in various departments, observe process flows and do modelling of the processes observed: Example: Spends time with the customer service team to understand the student enrollment challenges and workflows, then uses tools like UML diagrams and flowcharts to model software solutions that could automate and streamline enrollment processes and student management.	Created comprehensive process flow diagrams for student enrollment workflow, course management processes, and administrative functions using UML notation and flowchart tools.	18-Jan-2025	A. Makandla
WA0108	Understand management requirements and expectations from software development and software solutions: Example: Through meetings with the product management team, gain insights into the strategic goals for the student enrollment system, such as scalability (handling 5000+ students), performance benchmarks (<2 second page load), and user engagement metrics.	Documented management expectations: system must handle 5000+ concurrent users, maintain <2 second page load times, achieve 99.9% uptime, and provide comprehensive reporting capabilities.	20-Jan-2025	A. Makandla

Item	Scope	Comments	Date	Signature
WA0109	Understand business computer system and the support required from the software development team: Example: Learns about the existing student information systems, learning management systems, and identifies how the new enrollment system can integrate with existing infrastructure and databases to support institutional operations more effectively.	Analyzed existing systems architecture, documented integration requirements with current LMS, student information systems, and financial systems. Identified API requirements and database connectivity needs.	18-Jan-2025	A. Makandla
WA0110	Manage timesheets and apply self-management skills: Example: Use project management tools to track time spent on different development tasks, ensures you are prioritizing work according to project deadlines, and manages your development workload efficiently to maintain work-life balance while meeting sprint commitments.	Implemented time tracking using digital tools, logged 40 hours across various activities including development, learning, and documentation. Maintained detailed task breakdown and progress reports.	15-19 Jan 2025	A. Makandla
WA0111	Collaborate with team members throughout the work experience period to achieve common and individual goals: Example: Actively participates in agile sprints, collaborates through pair programming on PHP and JavaScript components, and contributes to the team's knowledge base by documenting solutions and troubleshooting steps for common web development problems.	Engaged in collaborative development sessions, participated in pair programming for frontend and backend components, contributed to team knowledge sharing through documentation and code reviews.	17-Jan-2025	A. Makandla

Item	Supporting Evidence	Evidenced Attached	Date	Signature
SE0101	Attendance Register	✓ [AttendanceRegister_Jan2025.pdf]	15-19 Jan 2025	A. Makandla
SE0102	Modelling	✓ [StudentEnrollmentFlow.pdf, ProcessModels.pdf]	18-Jan-2025	A. Makandla

WM-01-WE02: Observe and assist a software engineer with gathering of information and analysis of the requirements and problems

Scope of Work Experience

The person will be expected to engage in the following work activities:

- WA0201 Conduct an analysis of technical application requirements
- WA0202 Assist in the collection and documentation of user's requirements, development of user stories, and estimates
- WA0203 Conduct user research to identify either improvements to our existing services or where new services could be developed
- WA0204 Gather information from consumers about program functionality
- WA0205 Analyse user requirements
- WA0206 Analyse and identify requirements related to people, processes and technology
- WA0207 Identifying areas for modification in existing programs and subsequently developing these modifications
- WA0208 Convert requirements into design documents
- WA0209 The implementation of is assisted with
- WA0210 Conduct economic feasibility checks to ascertain that the project can be completed in the given budget
- WA0211 Conduct legal feasibility checks to ascertain that the project can be completed within regulatory compliances
- WA0212 Conduct operational feasibility checks to ascertain that the operations expected by the client can be created
- WA0213 Conduct technical feasibility checks to ascertain that the current computer system can support the software
- WA0214 Conduct scheduling feasibility checks to ascertain that the project can be completed within given schedule

Supporting Evidence

- SE0201 Analysis of the specification and problem

WM-01-WE02 OBSERVE AND ASSIST A SOFTWARE ENGINEER WITH GATHERING OF INFORMATION AND ANALYSIS OF THE REQUIREMENTS AND PROBLEMS

Item	Scope	Comments	Date	Signature
WA0201	Conduct an analysis of technical application requirements: Example:	Analyzed technical requirements for student enrollment system: LAMP	22- Jan-	A. Makandla

Item	Scope	Comments	Date	Signature
	Reviews a project brief for a student enrollment web application, assessing technical requirements like browser compatibility, server specifications (Apache/PHP/MySQL), responsive design needs, and security requirements for handling student data.	stack architecture, PHP 8.1 with MVC pattern, MySQL 8.0 database, responsive HTML5/CSS3 frontend with JavaScript ES6+, security requirements for student data protection.	2025	
WA0202	Assist in the collection and documentation of user's requirements, development of user stories, and estimates: Example: Participates in meetings with academic staff and administrators to gather requirements for the enrollment system. They help document these requirements in the form of user stories and contribute to estimating the development effort needed for PHP backend and JavaScript frontend components.	Created comprehensive user story backlog with 28 stories covering student registration, course enrollment, administrative functions, and reporting. Provided effort estimates using story points and time-based estimates for development sprints.	22-Jan-2025	A. Makandla
WA0203	Conduct user research to identify either improvements to our existing services or where new services could be developed: Example: Collaborates with the academic team to conduct surveys and user interviews with students and faculty to gather feedback on the current enrollment process, identifying pain points and potential features that could improve the user experience.	Conducted user research with 35 students and 12 faculty members through surveys and interviews. Identified key improvement areas: mobile-responsive interface (demanded by 89% of users), real-time enrollment status updates, and automated course prerequisite checking.	23-Jan-2025	A. Makandla
WA0204	Gather information from consumers about program functionality: Example: Sets up feedback sessions with current students who would use the enrollment system to understand their expectations for functionality, user interface preferences, and desired features for course selection and academic planning.	Organized focus group sessions with student representatives, gathered detailed feedback on desired functionality including course search/filter capabilities, schedule conflict detection, and mobile accessibility requirements.	24-Jan-2025	A. Makandla
WA0205	Analyse user requirements: Example: After collecting feedback from students and	Analyzed user feedback to prioritize core requirements: automated	24-Jan-	A. Makandla

Item	Scope	Comments	Date	Signature
	staff, analyzes user requirements to determine the technical specifications for the PHP backend, database schema design, and JavaScript frontend interactions needed to meet user expectations within the web application architecture.	enrollment validation, real-time course availability updates, student dashboard with progress tracking, and administrative reporting tools. Mapped requirements to technical implementation strategies.	2025	
WA0206	Analyse and identify requirements related to people, processes and technology: Example: Evaluates how the new enrollment system will impact current staff workflows, what training administrative staff will need, and whether the current server infrastructure can support the web application with expected user loads.	Conducted impact analysis covering staff training needs (estimated 16 hours), process changes in enrollment workflow, technology requirements including server capacity planning for 5000+ concurrent users, and database optimization strategies.	25-Jan-2025	A. Makandla
WA0207	Identifying areas for modification in existing programs and subsequently developing these modifications: Example: Identifies performance bottlenecks in existing student data systems, then develops optimized MySQL queries and improved PHP code structure to enhance data retrieval speed and overall system responsiveness.	Identified performance issues in legacy enrollment queries, developed optimized database indexes and improved PHP data access patterns, resulting in 65% improvement in page load times for student course listings.	26-Jan-2025	A. Makandla
WA0208	Convert requirements into design documents: Example: Translates detailed requirements for the enrollment system into comprehensive technical design documents including database ERD, PHP class diagrams, HTML wireframes, JavaScript component specifications, and system architecture documentation.	Created comprehensive design documentation including database ERD with 12 normalized tables, PHP MVC architecture specification, responsive HTML/CSS wireframes, JavaScript component diagrams, and API endpoint documentation.	27-Jan-2025	A. Makandla
WA0209	The implementation of is assisted with: Example: Assists in the development of enrollment system modules using PHP for backend logic, HTML/CSS for responsive frontend, JavaScript for interactive features, and MySQL for data persistence, ensuring proper integration between all components.	Assisted in implementing core enrollment modules: student registration (PHP/MySQL), course catalog (HTML/CSS/JavaScript), enrollment validation (PHP), and administrative dashboard components. Contributed to both frontend and backend development.	29-Jan-2025	A. Makandla

Item	Scope	Comments	Date	Signature
WA0210	Conduct economic feasibility checks to ascertain that the project can be completed in the given budget: Example: Evaluates development costs, hosting requirements, third-party service integrations, and ongoing maintenance expenses against the project budget to determine if the enrollment system can be delivered within financial constraints.	Completed comprehensive cost analysis: Development effort \$42,000, hosting infrastructure \$6,000, third-party integrations \$4,000, testing and deployment \$3,000. Total project cost \$55,000 within allocated budget of \$60,000.	27-Jan-2025	A. Makandla
WA0211	Conduct legal feasibility checks to ascertain that the project can be completed within regulatory compliances: Example: Reviews software compliance with educational data protection regulations (FERPA), web accessibility standards (WCAG 2.1), and data privacy laws to ensure the enrollment system meets all legal requirements for handling student information.	Conducted legal compliance review covering FERPA requirements for student data, GDPR compliance for international students, WCAG 2.1 AA accessibility standards, and institutional data governance policies. All requirements can be met within current framework.	27-Jan-2025	A. Makandla
WA0212	Conduct operational feasibility checks to ascertain that the operations expected by the client can be created: Example: Tests the proposed enrollment system functionality in simulated operational scenarios to verify it can handle peak enrollment periods, concurrent user loads, and integrate properly with existing academic systems.	Conducted operational feasibility testing with simulated enrollment scenarios. System successfully handled 1,500 concurrent users during peak enrollment simulation, maintained data integrity during high-load operations, and integrated smoothly with existing academic systems.	28-Jan-2025	A. Makandla
WA0213	Conduct technical feasibility checks to ascertain that the current computer system can support the software: Example: Conducts performance testing on current server infrastructure to evaluate whether it can support the PHP/MySQL enrollment system with expected user loads, and identifies any necessary hardware or software upgrades.	Evaluated current server infrastructure: Apache 2.4 with PHP 8.1 support confirmed, MySQL 8.0 compatibility verified, server capacity sufficient for initial deployment with auto-scaling capabilities for growth. Minor RAM upgrade recommended for optimal performance.	28-Jan-2025	A. Makandla

Item	Scope	Comments	Date	Signature
WA0214	Conduct scheduling feasibility checks to ascertain that the project can be completed within given schedule: Example: Assesses the project timeline considering development complexity, testing requirements, staff training, and deployment phases to ensure the enrollment system can realistically be completed within the academic calendar constraints.	Created detailed project timeline spanning 12 weeks: Requirements and design (2 weeks), Core development (6 weeks), Testing and integration (2 weeks), Training and deployment (2 weeks). Schedule feasible with current team capacity and allows buffer time for critical path activities.	28-Jan-2025	A. Makandla

Item	Supporting Evidence	Evidenced Attached	Date	Signature
SE0201	Analysis of the specification and problem	✓ [TechnicalRequirementsAnalysis_v2.1.pdf]	27-Jan-2025	A. Makandla

WM-01-WE03: Under supervision, gather information and analyse the requirements and problems by conducting the following tasks

Scope of Work Experience

The person will be expected to engage in the following work activities:

- WA0301 Conduct an analysis of technical application requirements
- WA0302 Assist in the collection and documentation of user's requirements, development of user stories, and estimates
- WA0303 Conduct user research to identify either improvements to our existing services or where new services could be developed
- WA0304 Gather information from consumers about program functionality
- WA0305 Analyse user requirements
- WA0306 Analyse and identify requirements related to people, processes and technology
- WA0307 Identifying areas for modification in existing programs and subsequently developing these modifications
- WA0308 Convert requirements into design documents
- WA0309 The implementation of is assisted with
- WA0310 Conduct economic feasibility checks to ascertain that the project can be completed in the given budget

- WA0311 Conduct legal feasibility checks to ascertain that the project can be completed within regulatory compliances
- WA0312 Conduct operational feasibility checks to ascertain that the operations expected by the client can be created
- WA0313 Conduct technical feasibility checks to ascertain that the current computer system can support the software
- WA0314 Conduct scheduling feasibility checks to ascertain that the project can be completed within given schedule

Supporting Evidence

- SE0301 Analysis of the specification and problem

WM-01-WE03 UNDER SUPERVISION, GATHER INFORMATION AND ANALYSE THE REQUIREMENTS AND PROBLEMS BY CONDUCTING THE FOLLOWING TASKS

Item	Scope	Comments	Date	Signature
WA0301	Conduct an analysis of technical application requirements: Reviews the technical specifications for the complete student enrollment system that integrates multiple components including PHP backend services, MySQL database optimization, responsive HTML5/CSS3 interface, and advanced JavaScript functionality for real-time updates and form validation.	Completed advanced technical analysis incorporating PHP 8.1 features (union types, attributes), MySQL 8.0 optimization with proper indexing strategies, CSS Grid and Flexbox for complex responsive layouts, and modern JavaScript (ES6+ modules, async/await, Fetch API) for enhanced user experience.	01-Feb-2025	A. Makandla
WA0302	Assist in the collection and documentation of user's requirements, development of user stories, and estimates: During advanced planning sessions, help gather complex requirements from multiple stakeholder groups including students, faculty, administrators, and IT staff for comprehensive enrollment management, translating these into detailed user stories with	Expanded user story collection to include 35 detailed stories covering advanced features: bulk enrollment operations, comprehensive reporting dashboard, notification systems, API integrations for third-party services. Refined estimates using velocity-based planning and risk assessment factors.	01-Feb-2025	A. Makandla

Item	Scope	Comments	Date	Signature
	acceptance criteria and refined development estimates.			
WA0303	Conduct user research to identify either improvements to our existing services or where new services could be developed: Participates in comprehensive user experience research including usability testing, A/B testing of interface designs, heat mapping analysis, and accessibility testing to identify opportunities for service enhancement and new feature development.	Conducted comprehensive UX research including usability testing with 25 participants, A/B testing of enrollment flow designs (achieving 23% improvement in completion rates), accessibility testing with screen readers, and mobile device testing across multiple platforms.	02-Feb-2025	A. Makandla
WA0304	Gather information from consumers about program functionality: Sets up comprehensive feedback collection systems including in-application feedback widgets, survey integration, user behavior analytics, and focus group sessions to gather detailed insights about desired functionality and user experience improvements.	Implemented comprehensive feedback collection system including embedded JavaScript feedback widgets, automated survey distribution, Google Analytics integration for user behavior tracking, and conducted detailed focus group sessions with quantitative analysis of results.	03-Feb-2025	A. Makandla
WA0305	Analyse user requirements: After gathering extensive feedback from diverse user groups, performs comprehensive analysis using requirement prioritization frameworks (MoSCoW method), user persona development, journey mapping, and technical feasibility assessment to create refined requirements specifications.	Performed advanced requirements analysis using MoSCoW prioritization, created 5 detailed user personas, developed user journey maps for different enrollment scenarios, and conducted technical feasibility assessment for each requirement with implementation complexity ratings.	03-Feb-2025	A. Makandla
WA0306	Analyse and identify requirements related to people, processes and technology: Examines comprehensive impact of	Conducted comprehensive impact analysis covering change management strategy for 45 staff members, developed 32-hour training curriculum with hands-on modules,	04-Feb-2025	A. Makandla

Item	Scope	Comments	Date	Signature
	the enrollment system implementation including change management requirements, staff training curricula development, process reengineering documentation, technology infrastructure assessment, and organizational readiness evaluation.	documented process reengineering for 7 departmental workflows, and assessed technology infrastructure scalability requirements.		
WA0307	Identifying areas for modification in existing programs and subsequently developing these modifications: Identifies comprehensive system optimization opportunities including database query optimization, PHP code refactoring for improved performance, CSS/JavaScript minification and bundling, caching strategies implementation, and security enhancements throughout the application stack.	Implemented comprehensive system optimizations: optimized 23 database queries (average 58% performance improvement), refactored PHP codebase using design patterns and dependency injection, implemented CSS/JavaScript minification reducing bundle size by 42%, and added Redis caching layer improving response times by 71%.	05-Feb-2025	A. Makandla
WA0308	Convert requirements into design documents: Takes finalized comprehensive requirements and creates detailed technical design documentation including complete system architecture diagrams, detailed database schema with relationships and constraints, comprehensive API documentation, UI/UX design specifications, and security architecture documentation.	Created comprehensive design documentation suite including: complete system architecture with microservices design, normalized database schema with 15 tables and comprehensive relationship mapping, RESTful API documentation with OpenAPI specification, detailed UI/UX mockups with interactive prototypes, and security architecture with authentication/authorization flows.	06-Feb-2025	A. Makandla
WA0309	The implementation of is assisted with: Assists in comprehensive system implementation including full-stack development using PHP MVC framework, responsive frontend development with	Contributed to full-stack implementation: developed 12 PHP controller classes with service layer architecture, created responsive frontend with 18 UI components, implemented database layer with stored procedures and triggers, developed	08-Feb-2025	A. Makandla

Item	Scope	Comments	Date	Signature
	progressive enhancement, database implementation with advanced features, JavaScript application development, testing framework implementation, and deployment automation.	JavaScript modules for real-time features, created comprehensive test suite with 85% code coverage, and automated deployment pipeline.		
WA0310	Conduct economic feasibility checks to ascertain that the project can be completed in the given budget: Reviews comprehensive cost analysis including detailed development effort estimates, infrastructure costs with scaling considerations, third-party service costs, ongoing maintenance and support costs, and ROI projections to ensure project viability within budget constraints.	Completed detailed economic analysis: Development costs \$52,000 (including advanced features), infrastructure costs \$8,000 (with auto-scaling), third-party services \$5,000, training and change management \$7,000, ongoing maintenance \$12,000/year. Total implementation cost \$72,000 with 3-year ROI of 245% through efficiency gains.	06-Feb-2025	A. Makandla
WA0311	Conduct legal feasibility checks to ascertain that the project can be completed within regulatory compliances: Consults with legal and compliance experts to ensure comprehensive adherence to educational regulations (FERPA), international privacy laws (GDPR), accessibility compliance (WCAG 2.1 AA), data sovereignty requirements, and institutional governance policies throughout the system design and implementation.	Conducted comprehensive legal compliance review: FERPA compliance framework implemented with role-based access controls, GDPR compliance with consent management and data portability features, WCAG 2.1 AA accessibility achieved through semantic HTML		